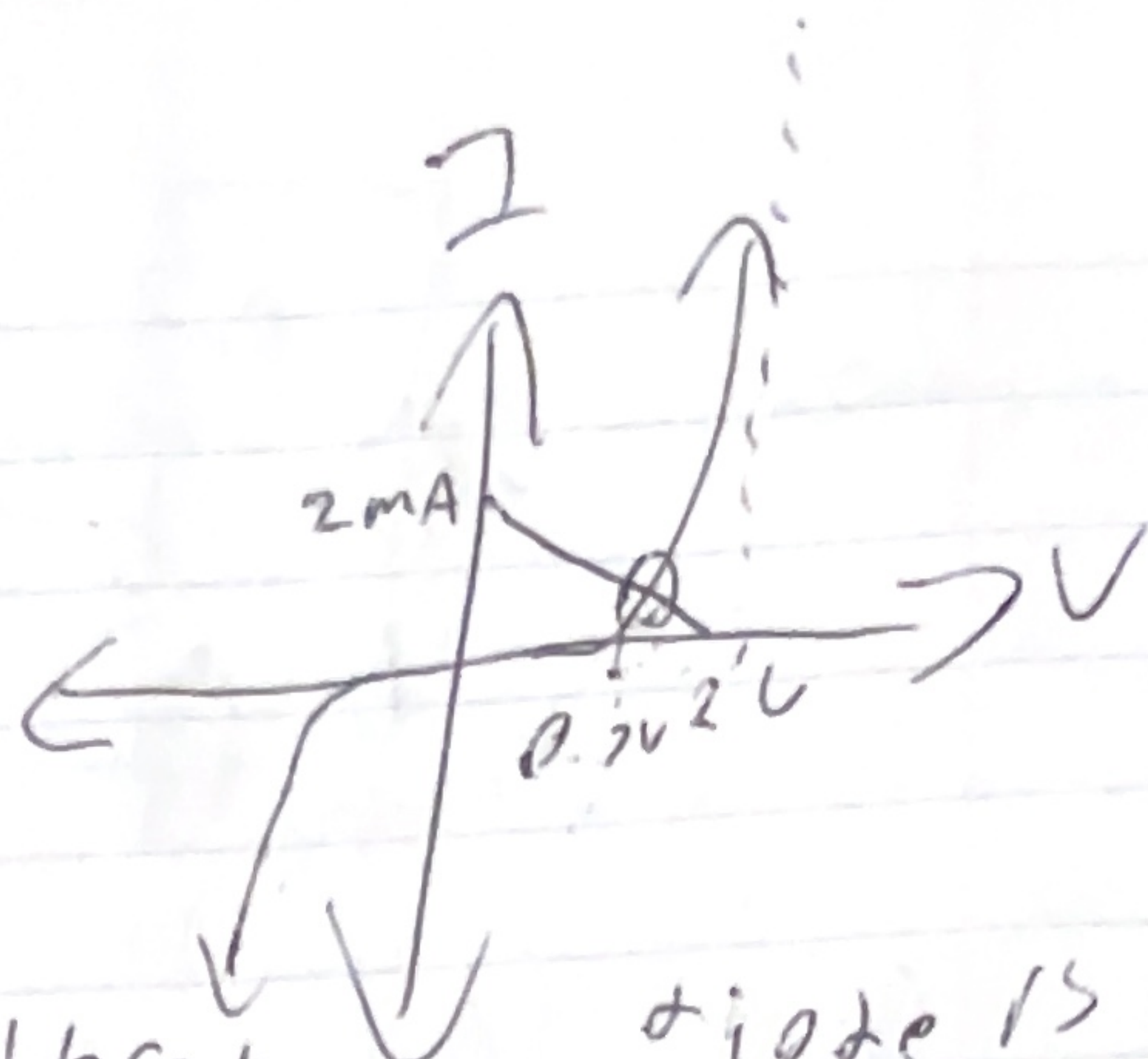
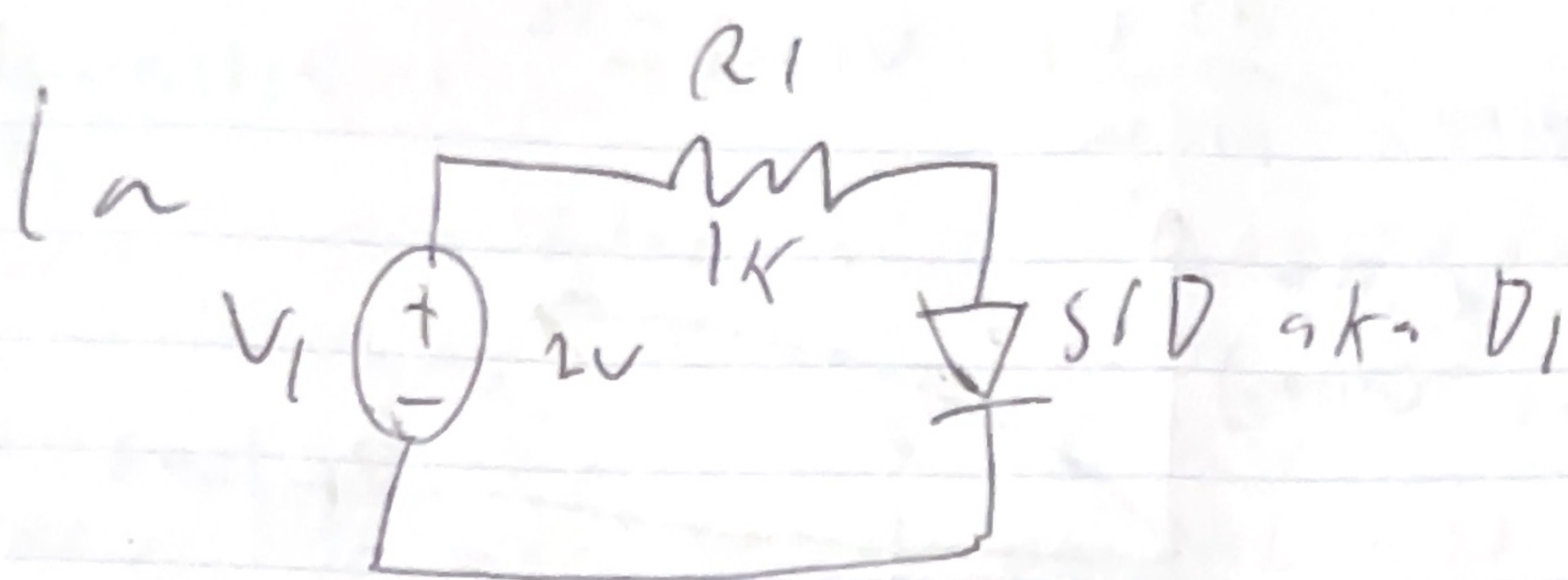


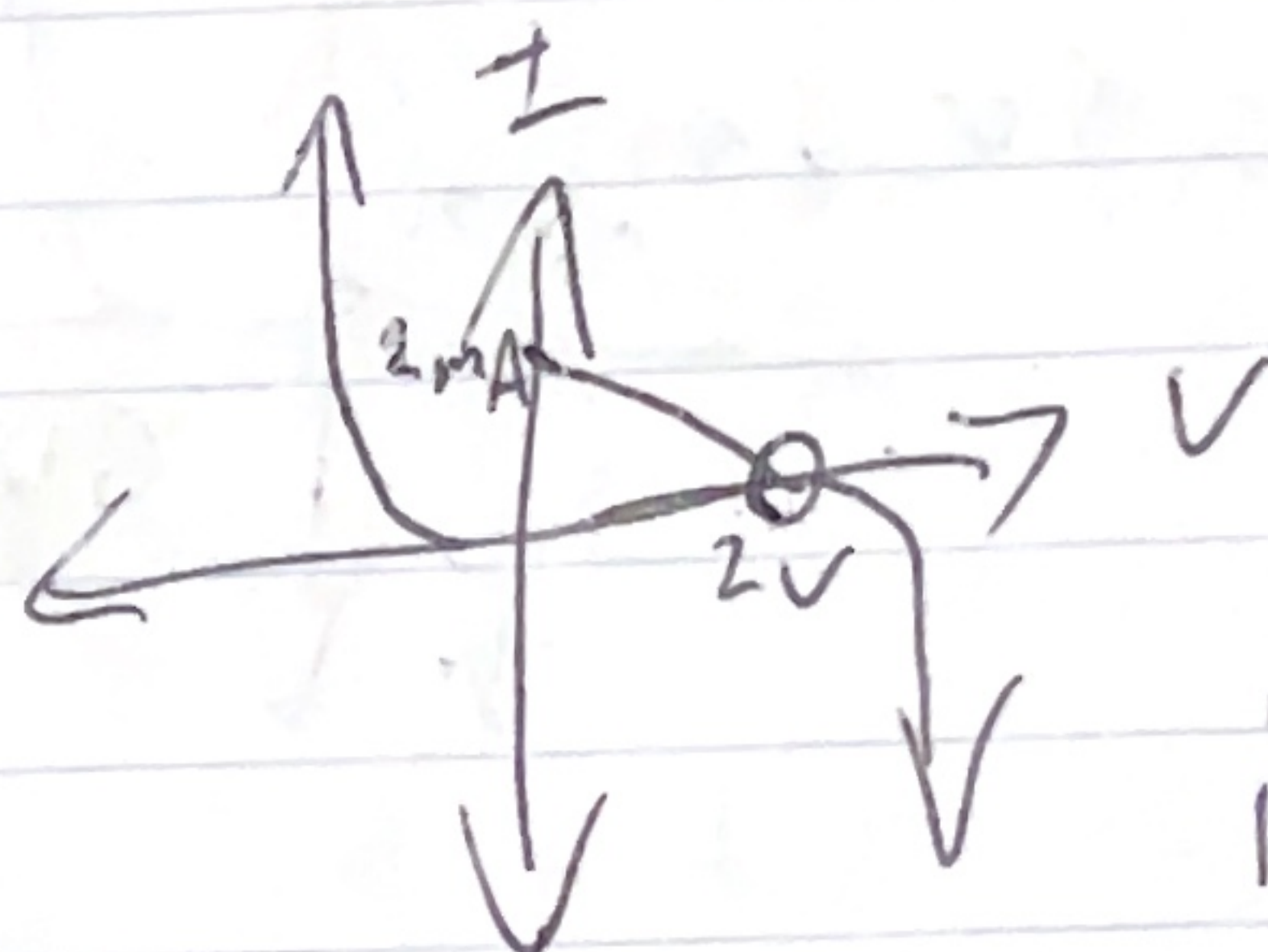
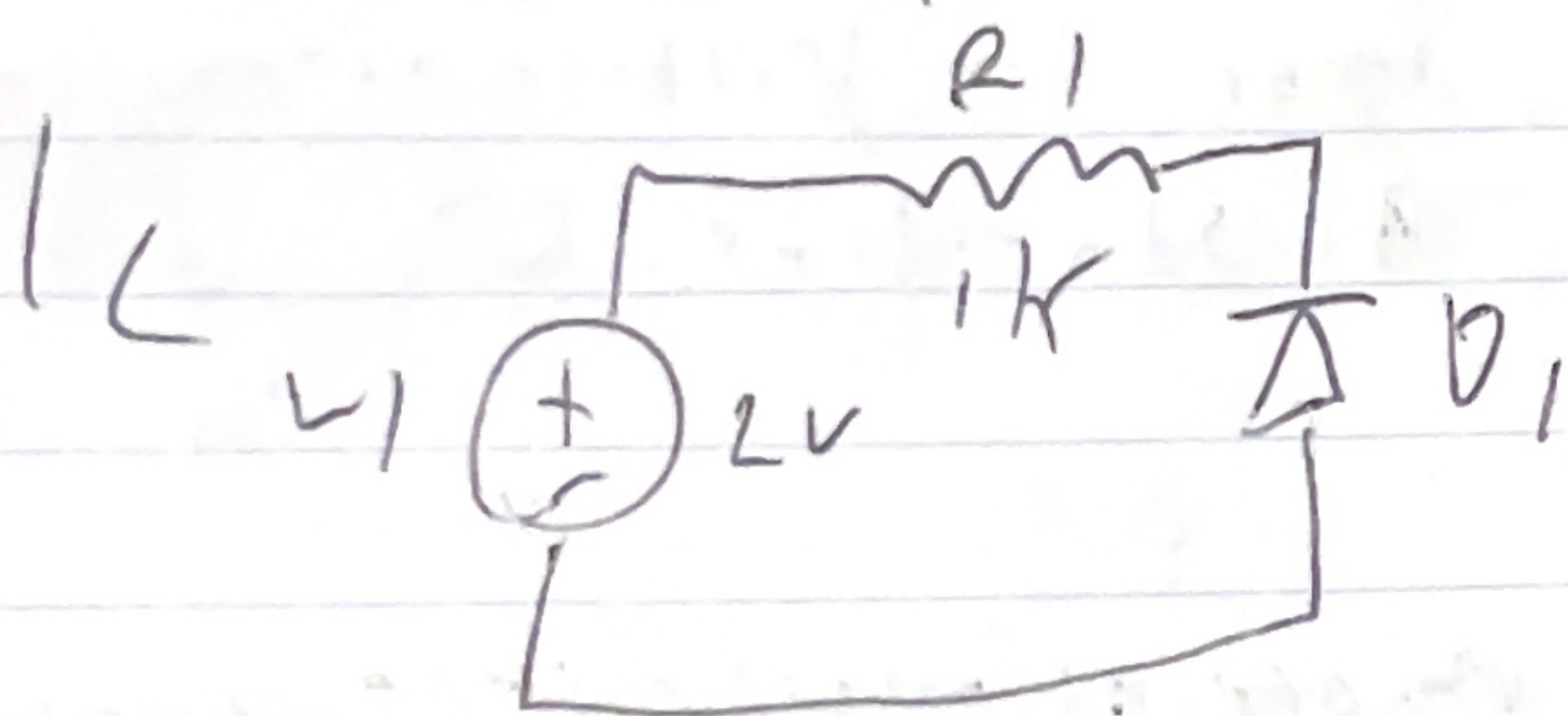


With 2V power source, current through diode is very high (lets all current through). Probably in range of 10^6 or 10^7 ,

1b $I = I_0 (e^{V/V_T})$ $I_0 = 10^{-9}$ $V_T = 26mV$

$$I = 10^{-9} (e^{2/0.026}) \rightarrow 2.55 \times 10^{24} A$$

Answers agree, both are very large numbers so they agree



Diode current is roughly 0A

1.2 voltage is flipped, so equation becomes

$$I = 10^{-9} (e^{-2/0.026}) = 3.92 \times 10^{-43} A$$

Numbers agree!

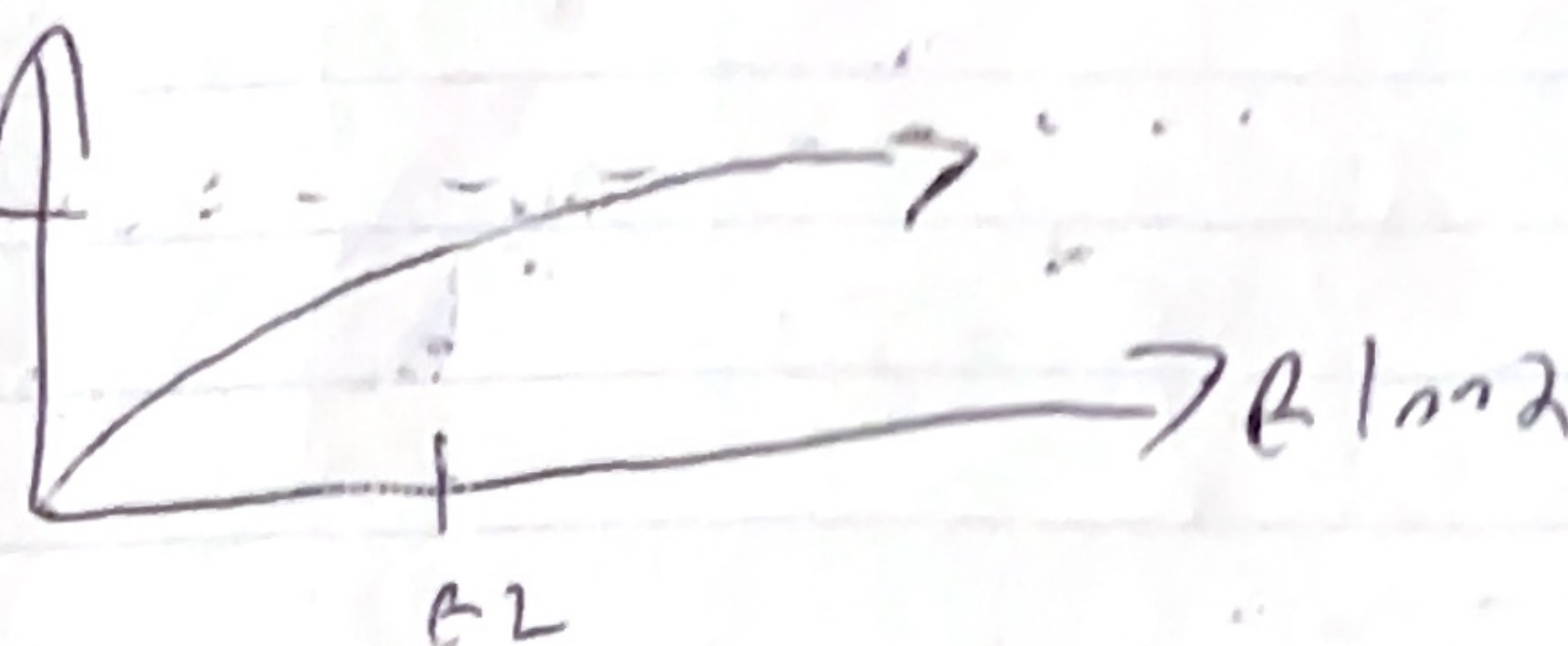
2a



$$V_{load} = \frac{R_2 V_1}{R_1 + R_2}$$

$$V_{load} = V_L \frac{R_2 || R_{load}}{R_1 + (R_2 || R_{load})}$$

$$\frac{V_1 \cdot R_2}{(R_1 + R_2)}$$



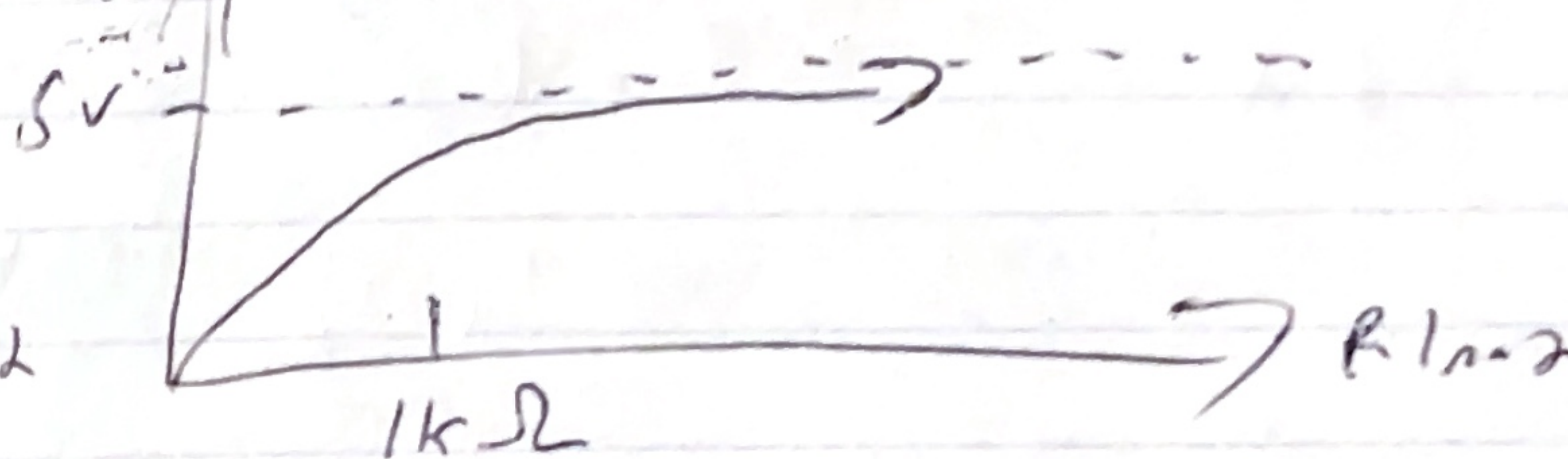
2b

R_{load} must be much greater than other resistances otherwise current will go through load instead of a circuit.

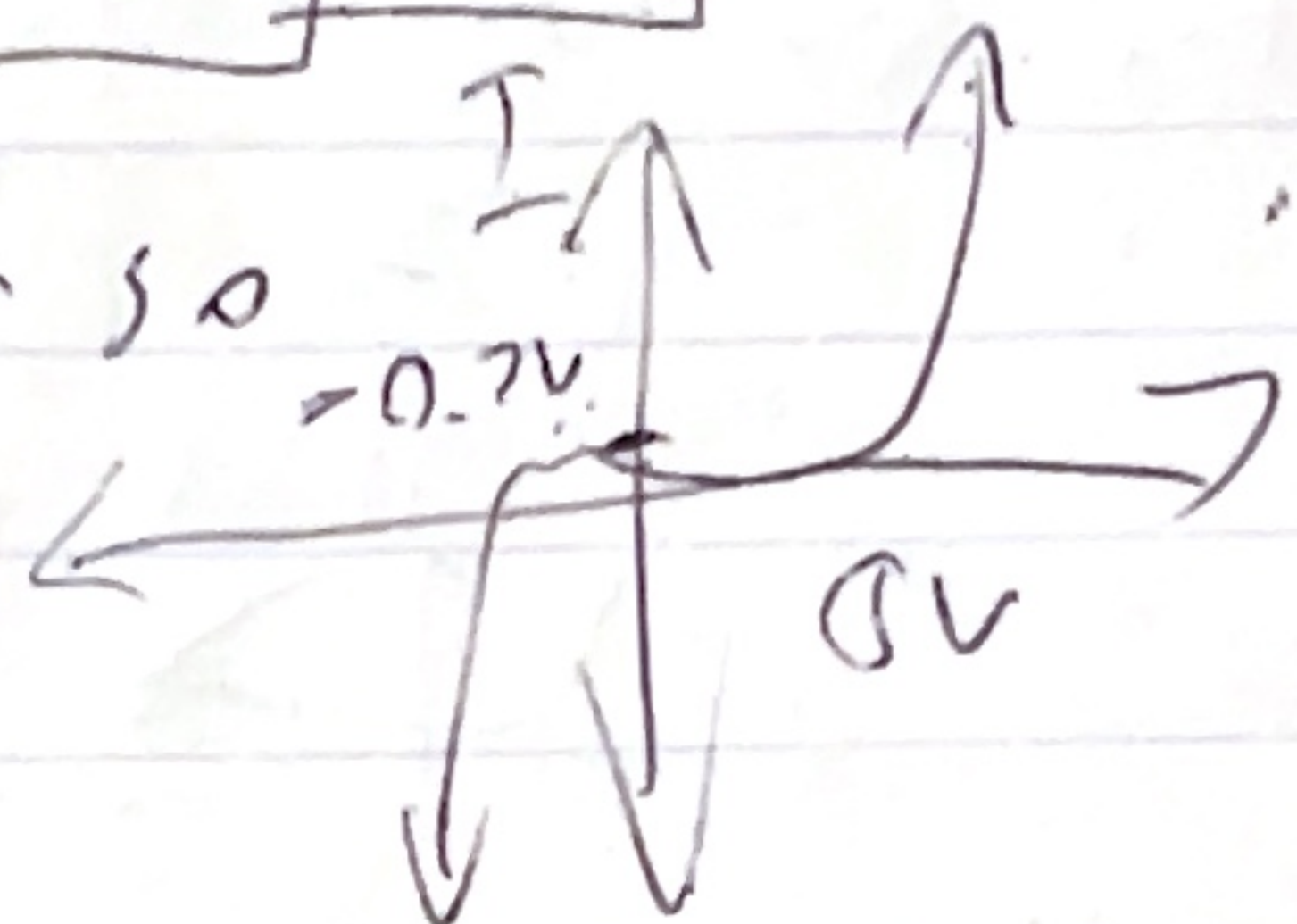
2c



V_{load}



$V_2 = 5V$ so



2d Same, R_{load} must be large for Voltage across load to stabilize.

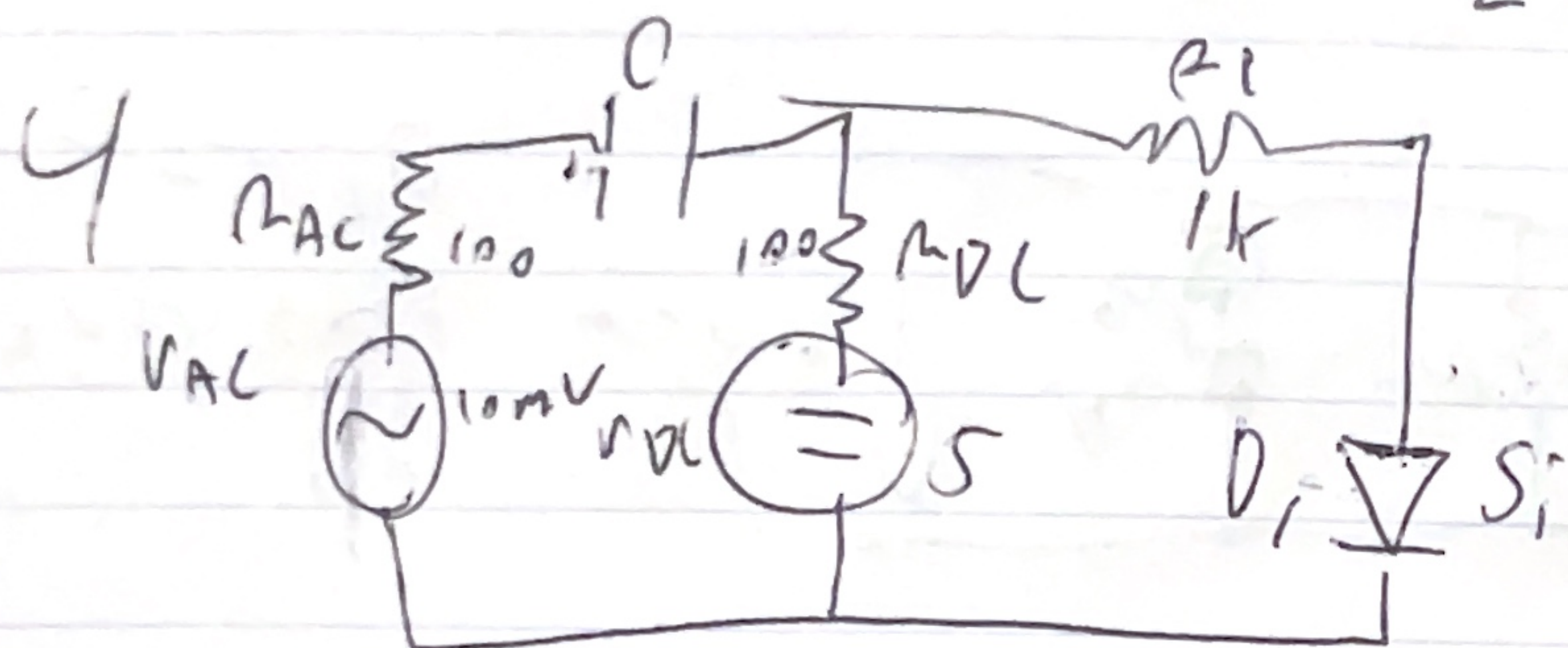
2e

First circuit has a more adjustable curr since switching R_1 will result in V_{load} changing more. But second circuit, thanks to Zener diode, has a much more adjustable voltage stabilization, as it is equal to V_2 .

$V_{\text{source}} = V_{\text{mid}} \sim V_2$

3a The Zener diode is there to smooth the voltage out.

3b A large part of the AC voltage is clipped off if V_A of V_{source} is $> V_2$.



4a V_{DC} sees 1100 Ω resistance. R_{DC} and R_1 in series since C lots through no DC.

$$I = \frac{V_{DC} - V_{\text{diode}}}{R} = \frac{4.3}{1100} = 3.9 \text{ mA} \approx I_{DC}$$

4b $r_2 = \frac{v_t}{I_{DC}} = \frac{26 \text{ mV}}{3.9 \text{ mA}} = 6.66 \Omega$

Draw AC = shorts DC so C can be shorted too with equivalent sources. R_{ACDC} becomes 50 and $v_{AC} \rightarrow 20 \text{ mV}$



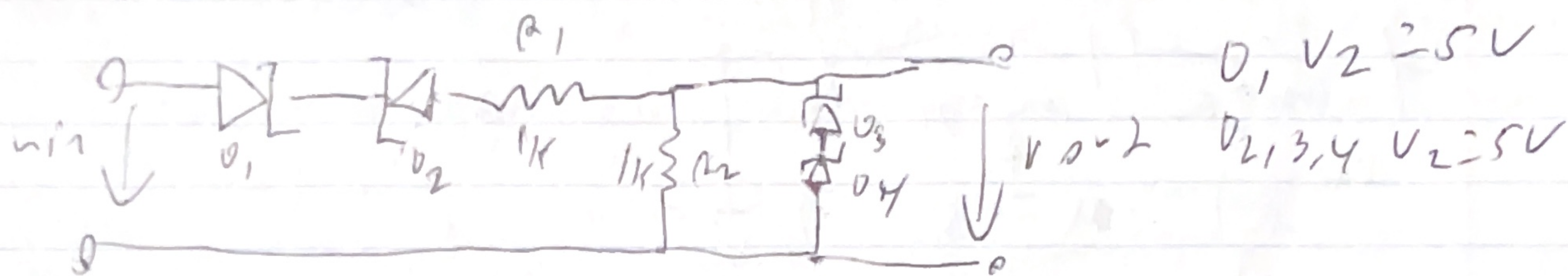
$$4c \quad V_{AC}^{diode} = \frac{r_2}{R_T + r_2} V_{AC}$$

$$r_2 = 6.66 \quad R_T = 1050 \quad V_{AC} = 20 \text{ mV}$$

$$V_{AC}^{diode} = 0.13 \text{ mV}$$

$$V = I R \quad V = 0.13 \quad R = 6.66 \quad I = 0.0195 \text{ mA}$$

5a Diode threshold voltage is the voltage at which the diode has enough energy to "turn on". There are silicon Zener diodes so the value for the circuit is $-0.7V$. Zener voltage is the voltage where reverse breakdown happens, which allows current through. In the circuit it is $5V$.

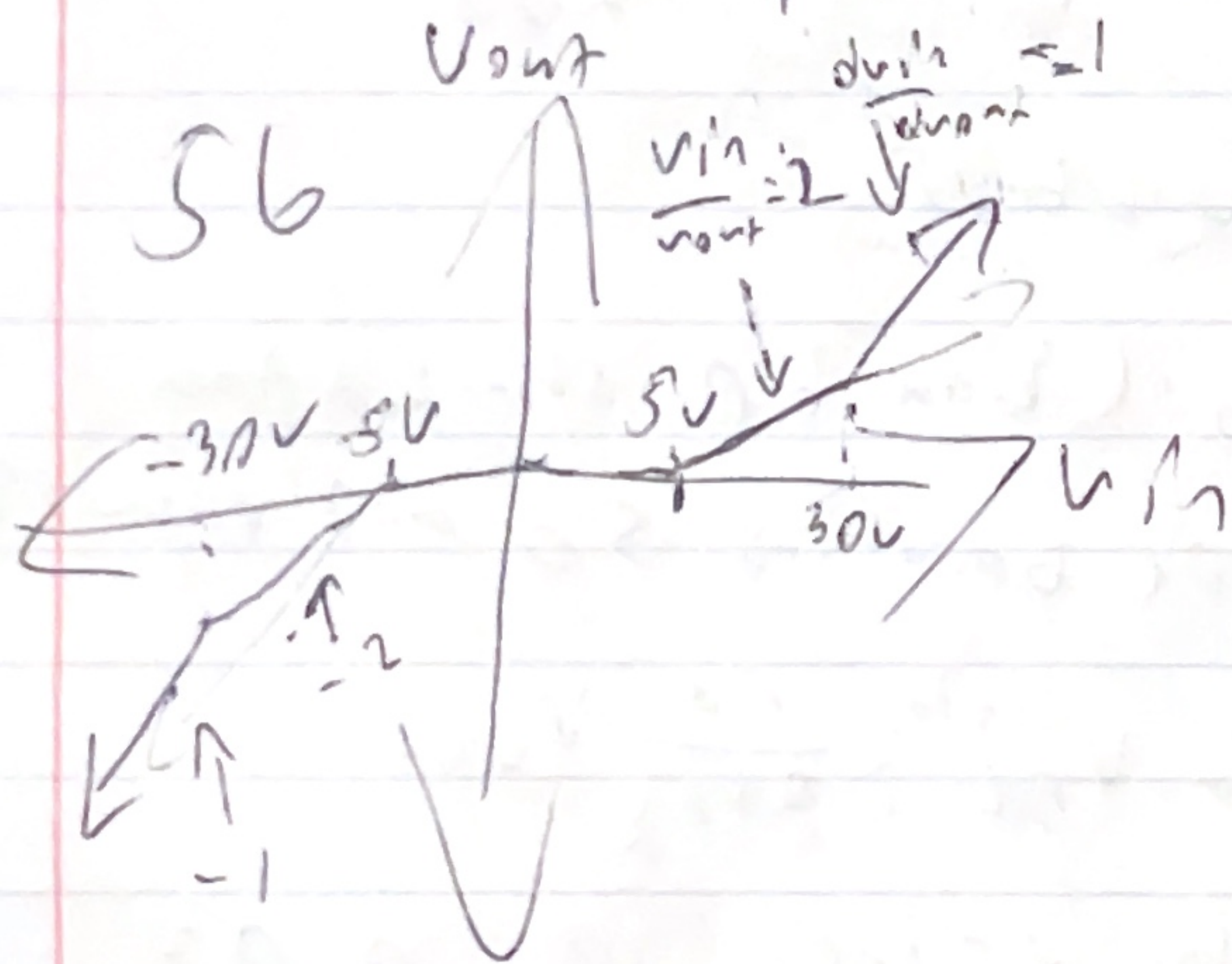


D_1 and D_2 make a reverse clipper.

R_1 and R_2 make a $\frac{1}{2}$ voltage divider.

D_3 and D_4 smooth (no real effect on DC)

EXCEPT at $10V$ across $D_3 D_4$ where voltage divider becomes absolute



5c There is no voltage between $-5V$ and $5V$ because of the reverse clipper D_1 and D_2 . They only let voltage through if $|V| > V_Z$. There is a slope of $\frac{1}{2}$ when $|V| < 5$ and $|V| > 5$ and a slope of 1 when $|V| > 30$. This is because

D_1 and D_2 steal $10V$ s, R_1 and R_2 halve the voltage, and it takes $10V$ going across $D_3 D_4$ for voltage to go through.

$$\frac{1}{2}(X-10) = 10 \Rightarrow X = 30$$

$$52 \frac{1}{2}(X-5) \rightarrow \frac{1}{2} \cdot 5 = \left(\frac{5}{2}V \text{ and } \frac{5}{2}mA \right)$$

$$5e \quad X = 17.5 \Rightarrow (12.5V \text{ and } 12.5mA)$$